

Integration of Network Functions Virtualization Infrastructure with the DevOps Pipeline

This series' eighth and final article outlines the creation of a DevOps pipeline to deploy a network functions virtualization module. This pipeline takes care of all the challenges faced from development to deployment of software, making the process simpler for developers and operations teams.



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DevOps has been known to reduce deployment time, and its automation chain can help push the latest application features and fixes in no time. So how about pushing virtualised network functions through this DevOps pipeline on a platform that could help scale the application both horizontally and vertically across a cloud computing platform?

In this article, we have successfully created a pipeline that achieves all the benefits of network functions virtualization (NFV) while easing the burden on developers and operations teams when implementing the DevOps pipeline. This DevOps pipeline takes care of every single hurdle, from development to deployment. The current setup utilises cutting-edge

technologies such as distributed systems, containerisation, software-defined networking, and cloud computing, all under the umbrella of DevOps. This article demonstrates how to deploy an NFV module such as a video-on-demand (VOD) containerised application over an NFV-enabled infrastructure using a DevOps pipeline.

Experimental setup

We have configured a pipeline that invokes various DevOps tools in different phases, where the end goal is to achieve full-fledged deployment of our NFV application. The integration is carefully done to fulfil each stage of DevOps, invoke specific automation tools to accomplish the task, and then call the subsequent phases.

The source code is developed by the developer and is pushed to a source control management (SCM) system. With every push to the SCM, the integration tool is invoked to run the predefined tasks while containerising the microservices of the developed application. This container image is then deployed over the software-defined networking (SDN)-enabled cloud platform, which also orchestrates all the running containers. Let's explore how all these tools work in sync for faster deployment of the application.

Managing code with GitHub

GitHub is an excellent SCM tool built by the open source community for developers. It is utilised to manage the various code versions and branches.